



Introduction to Photogrammetry

Lecture 15

Exterior Orientation

Usman Mazhar, Ph.D.

usmanmazhargeo@outlook.com

In Today's Class

1. Introduction
2. Exterior Orientation Parameters
3. Coordinate Systems in Exterior Orientation
4. Methods of Exterior Orientation
5. Sources of Error in Exterior Orientation

1. Introduction

- Exterior orientation determines the position and angular orientation of the camera during image exposure.
- The process reconstructs the geometry between the photograph and the ground surface.
- Exterior orientation allows image measurements to be transformed into real-world coordinates.
- It is an essential stage in analytical photogrammetry and spatial reconstruction

- Exterior orientation establishes the spatial relationship between photographs and terrain.
- The process enables accurate determination of object coordinates from image coordinates.
- Exterior orientation supports topographic mapping and three-dimensional modeling.
- Accurate orientation improves the geometric reliability of photogrammetric products

- Exterior orientation enables accurate terrain reconstruction from photographs.
- The process supports digital elevation model generation.
- Exterior orientation is essential for map compilation and orthophoto production.
- Many remote sensing and GIS applications depend on accurate orientation.

- Exterior orientation is used in aerial surveying and topographic mapping.
- The method supports drone photogrammetry and UAV mapping systems.
- It is applied in engineering surveys and infrastructure planning.
- Exterior orientation is also important in satellite photogrammetry.

- GPS and IMU systems now support automated exterior orientation.
- Digital photogrammetry has improved computational efficiency significantly.
- Automated tie point extraction reduces manual processing requirements.
- Modern software performs orientation procedures rapidly and accurately.

Relationship Between Interior and Exterior Orientation

- Interior orientation reconstructs the internal geometry of the camera system.
- Exterior orientation determines the camera position and attitude in object space.
- Both orientation procedures are required for complete photogrammetric reconstruction.
- Exterior orientation cannot be performed accurately without proper interior orientation

Concept of Perspective Geometry

- Photogrammetry is based on the principle of central perspective projection.
- Ground points are projected through the camera lens onto the image plane.
- The perspective center represents the location of the camera exposure station.
- Exterior orientation reconstructs this perspective relationship mathematically.

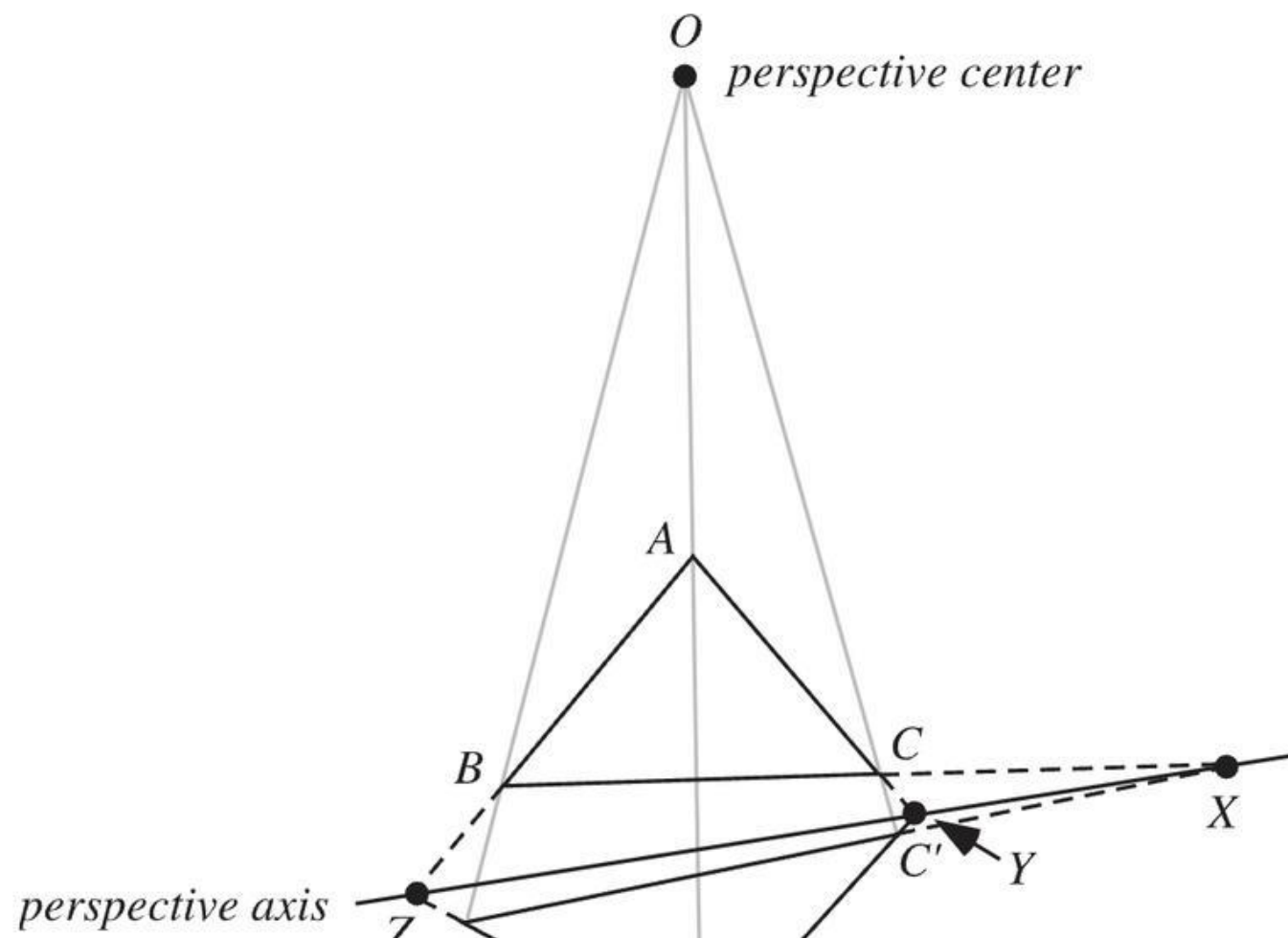
2. Exterior Orientation Parameters

- Exterior orientation is defined by six parameters describing camera position and rotation.
- Three parameters represent the camera location in object space.
- Three parameters represent the angular orientation of the camera axes.
- These parameters collectively define the imaging geometry.

Position Parameters

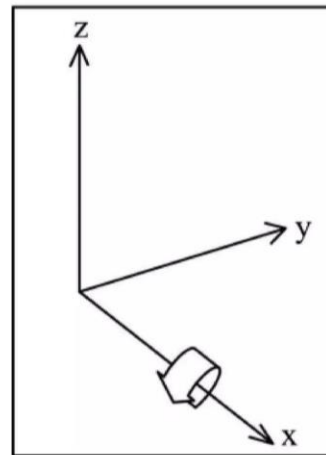
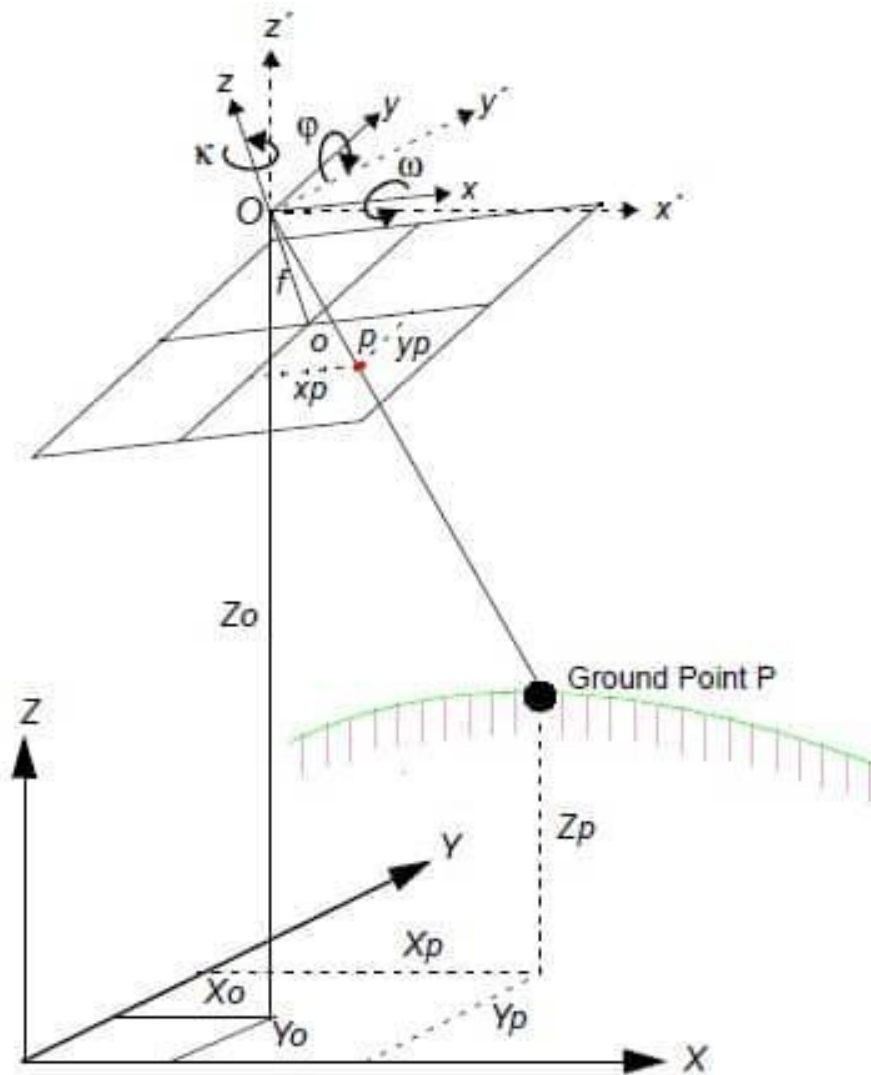
- The camera position is represented by X_0 , Y_0 , and Z_0 coordinates.
- These coordinates define the location of the perspective center in ground space.
- Position parameters determine where the photograph was taken.
- Accurate position determination is important for spatial accuracy

Perspective Center

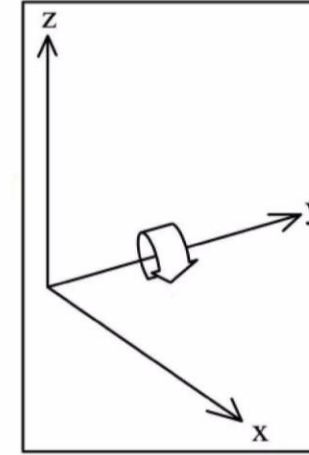


Rotation Parameters

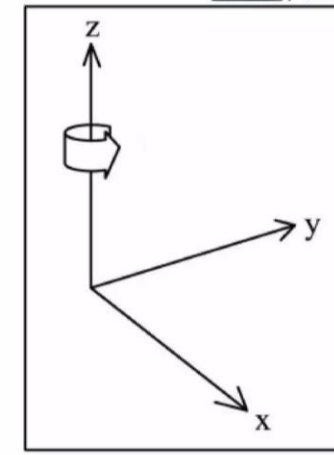
- Rotation parameters describe the angular orientation of the camera.
- Omega (ω) represents rotation around the x-axis.
- Phi (φ) represents rotation around the y-axis.
- Kappa (κ) represents rotation around the z-axis.



Rotation around the X-axis (ω)



Rotation around the Y-axis (ϕ)



Rotation around the Z-axis (κ)

- **Omega** (ω) is the rotation around the X-axis.
- **Phi** (ϕ) is the rotation around the Y-axis.
- **Kappa** (κ) is the rotation around the Z-axis.
- The omega, phi, kappa are in **degrees** and can be positive or negative. This means, for example, that the value 350 degrees can be expressed also as -10 [0-(360-350)].

Omega Rotation

- Omega rotation occurs around the x-axis of the photograph.
- This rotation tilts the photograph forward or backward.
- Omega influences the vertical orientation of the image.
- Accurate omega values are necessary for proper image alignment

Phi Rotation

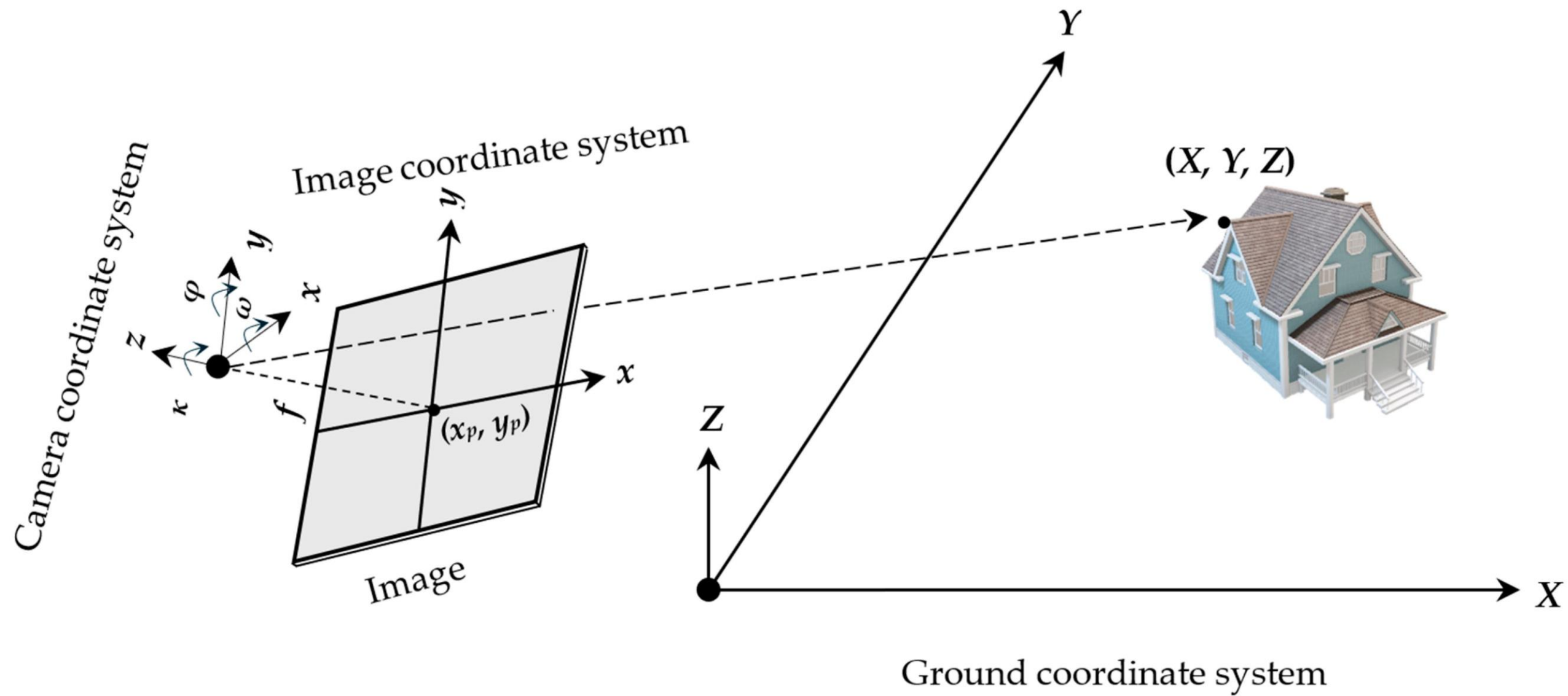
- Phi rotation occurs around the y-axis of the photograph.
- This rotation tilts the image sideways.
- Phi affects the lateral orientation of the camera system.
- Proper phi correction improves geometric consistency.

Kappa Rotation

- Kappa rotation occurs around the z-axis of the photograph.
- This rotation changes the horizontal orientation of the image.
- Kappa is associated with image rotation within the image plane.
- Correct kappa orientation is necessary for map alignment.

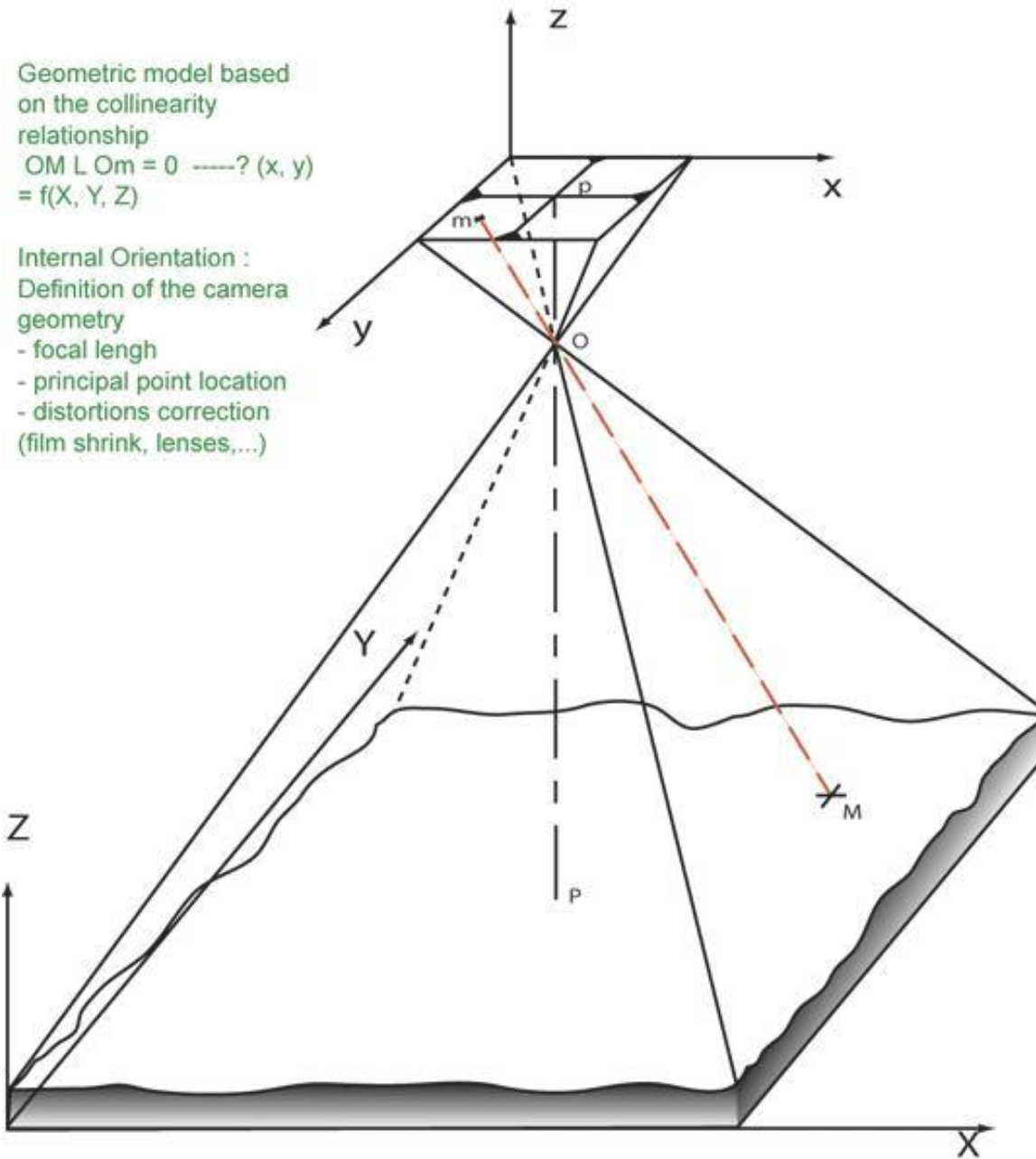
3. Coordinate Systems in Exterior Orientation

- Exterior orientation involves both image and ground coordinate systems.
- Image coordinates are measured on the photograph.
- Ground coordinates represent real-world object locations.
- Mathematical relationships connect these coordinate systems.



Collinearity Principle

- The perspective center, image point, and ground point lie on the same straight line.
- This condition is known as the collinearity principle.
- Collinearity forms the mathematical foundation of exterior orientation.
- The principle is expressed through collinearity equations.

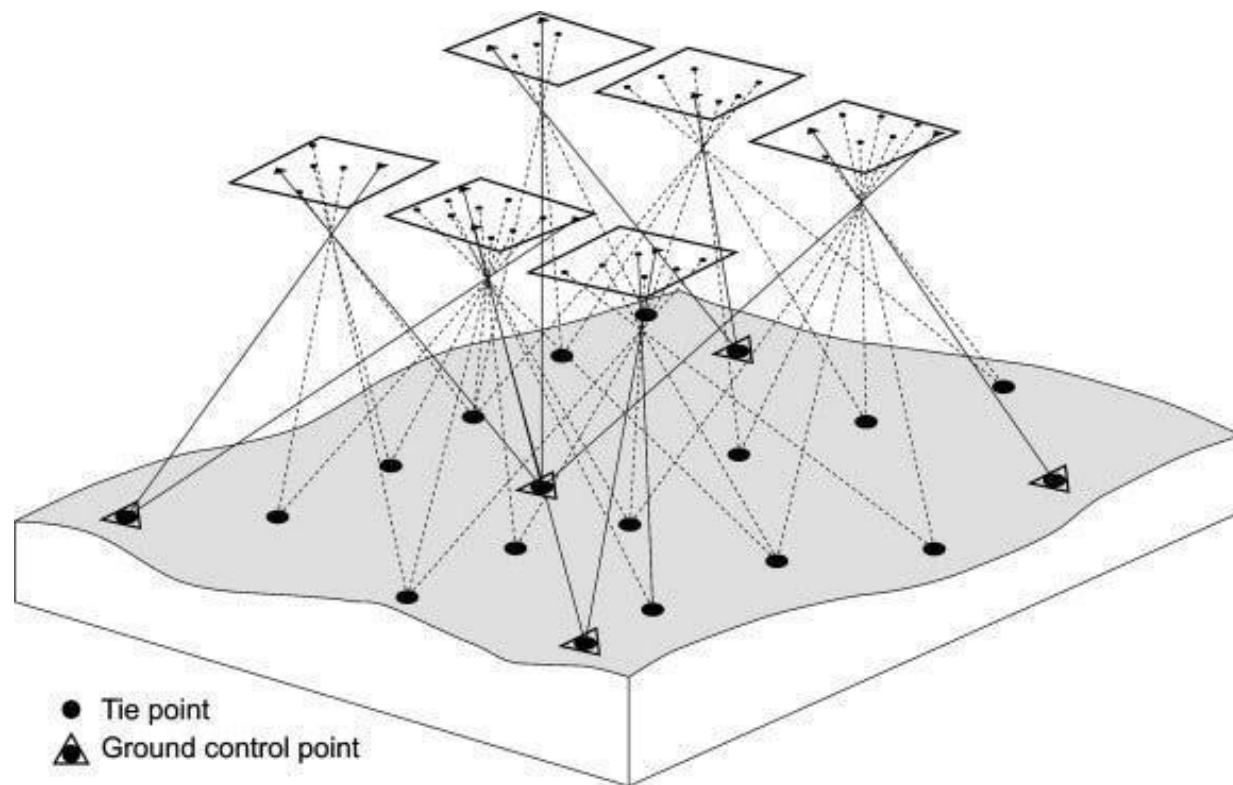


Collinearity Equations

- Collinearity equations relate image coordinates to ground coordinates mathematically.
- The equations include camera position and rotation parameters.
- These equations are used to compute object coordinates from photographs.
- Collinearity equations are fundamental in analytical photogrammetry.

Ground Control Points

- Ground control points are locations with known ground coordinates.
- These points are used to determine exterior orientation parameters.
- Ground control points connect image space with object space.
- Accurate control points improve orientation precision.



4. Methods of Exterior Orientation

- Exterior orientation can be performed using analytical or numerical methods.
- Ground control points are commonly used in orientation procedures.
- Bundle adjustment techniques improve orientation accuracy.
- Modern systems often use automated orientation algorithms.

Space Resection

- Space resection determines camera orientation from known ground control points.
- The method computes camera position and rotation parameters simultaneously.
- Space resection is widely used in aerial triangulation.
- At least three control points are generally required.

5. Sources of Error in Exterior Orientation

- Errors in image measurements can affect orientation accuracy.
- Inaccurate ground control points may distort spatial reconstruction.
- Lens distortions and atmospheric effects may introduce additional errors.
- Proper adjustment procedures help reduce orientation uncertainties.



Thank you!

Any Questions
